

$$\textcircled{1} a) \nabla \times F = \begin{vmatrix} e_1 & e_2 & e_3 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x - \alpha z e^{2x} & 3y^2 & z^2 - e^{2x} \end{vmatrix}$$

$$= (0 - 0, -\alpha e^{2x} + 2e^{2x}, 0 - 0) = (0, 0, 0) \text{ na } \mathbb{R}^3 \Leftrightarrow$$

$$\Leftrightarrow \alpha = 2.$$

$$b) \frac{\partial f}{\partial x} = x - 2ze^{2x} \quad (1)$$

$$\frac{\partial f}{\partial y} = 3y^2 \quad (2)$$

$$\frac{\partial f}{\partial z} = z^2 - e^{2x} \quad (3)$$

$$(1) \Rightarrow f(x, y, z) = \frac{x^2}{2} - ze^{2x} + \tilde{C}(y, z)$$

Dodavajmo do (2) našme $\frac{\partial \tilde{C}}{\partial y} = 3y^2 \Rightarrow \tilde{C}(y, z) = y^3 + l(z)$

Z (3) polo plyneme, že

$$-e^{2x} + C'(z) = z^2 - e^{2x}$$

Proto $C(z) = \frac{z^3}{3} + K, K \in \mathbb{R}$

Z $f(0, 0, 1) = 0$ dedujeme $0 = -1 + \frac{1}{3} + K$. Odhad

$$f(x, y, z) = \frac{x^2}{2} - ze^{2x} + y^3 + \frac{z^3}{3} + \frac{2}{3}$$

c) Pro $\alpha = 2$ je F potenciálové na $\mathbb{R}^3$

a $C \subseteq \mathbb{R}^3$ je uzavřená křivka. Proto $\int_C F = 0$.

2a) Hledáme body minima funkce

$$f(x, y) = (-2x + y + 1)^2 + (y - 1)^2 + (2x + y - 2)^2$$

na množině $H = \mathbb{R}^2$.

b) Protože f je konvexní, je každý stacionární bod bodem minima.

$$\frac{\partial f}{\partial x} = -4(-2x + y + 1) + 4(2x + y - 2) = 0 \quad (1)$$

$$\frac{\partial f}{\partial y} = 2(-2x + y + 1) + 2(y - 1) + 2(2x + y - 2) = 0 \quad (2)$$

$$(1) \Leftrightarrow 4x - 3 = 0 \Leftrightarrow x = \frac{3}{4}$$

$$(2) \Leftrightarrow 3y - 2 = 0 \Leftrightarrow y = \frac{2}{3}$$

Hledaný příklad má proto rovnici $y = \frac{3}{4}x + \frac{2}{3}$.

Alternativně můžeme v řešení použít maticový zápis úlohy podle s teorému z přednášky:
minimální řešení $f(x, y) = \|A \begin{pmatrix} x \\ y \end{pmatrix} - b\|$ na $\mathbb{R}^2$, kde

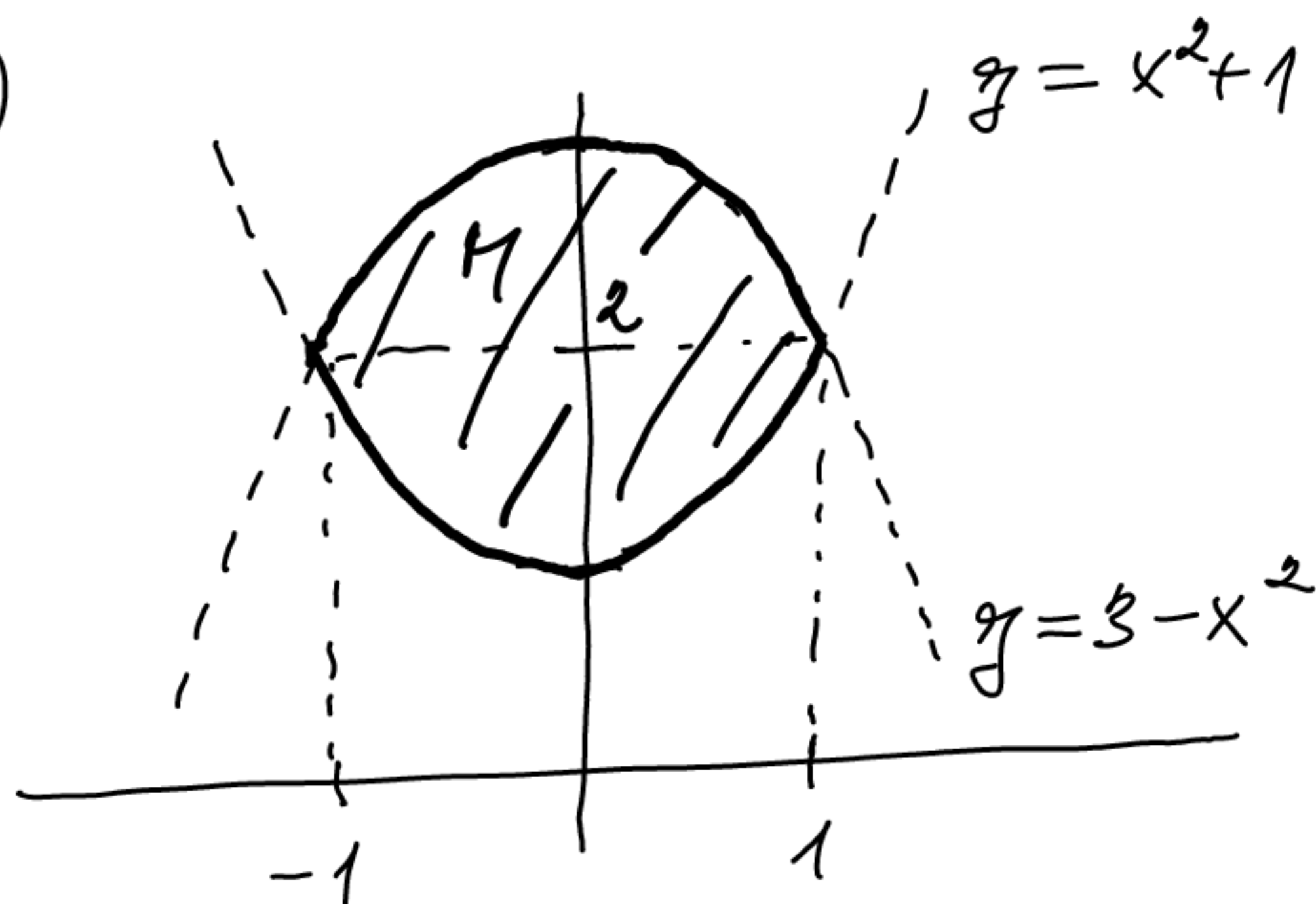
$$A = \begin{pmatrix} -2 & 1 \\ 0 & 1 \\ 2 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}.$$

Protože $A^T A = \begin{pmatrix} 8 & 0 \\ 0 & 3 \end{pmatrix}$, je $(A^T A)^{-1} = \frac{1}{24} \begin{pmatrix} 3 & 0 \\ 0 & 8 \end{pmatrix}$.

Odtud

$$\begin{pmatrix} x \\ y \end{pmatrix} = (A^T A)^{-1} A^T b = \frac{1}{24} \begin{pmatrix} 3 & 0 \\ 0 & 8 \end{pmatrix} \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \frac{1}{24} \begin{pmatrix} 18 \\ 16 \end{pmatrix} = \begin{pmatrix} \frac{3}{4} \\ \frac{2}{3} \end{pmatrix}.$$

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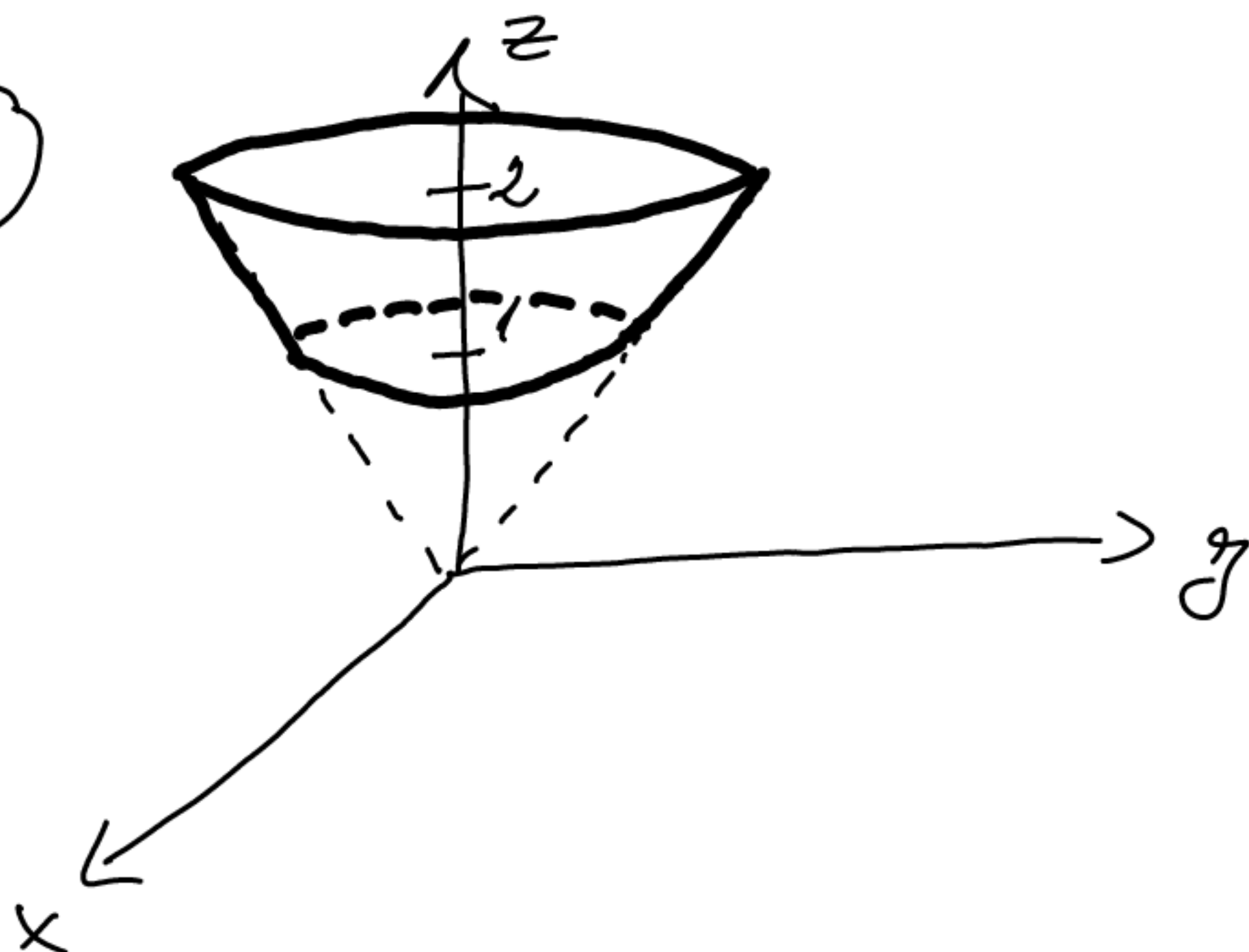
$$\int_{\Omega} y = \int_{-1}^1 \int_{x^2+1}^{3-x^2} y \, dy \, dx = \int_{-1}^1 \left[\frac{y^2}{2} \right]_{x^2+1}^{3-x^2} dx$$

$$= \frac{1}{2} \int_{-1}^1 (3-x^2)^2 - (x^2+1)^2 dx$$

$$9 - 6x^2 + x^4 - (x^4 + 2x^2 + 1) = 8(1-x^2)$$

$$= 8 \int_0^1 1-x^2 dx = 8 \left(1 - \frac{1}{3} \right) = \frac{16}{3}.$$

④



$$D \cdot F = y^2 + x^2$$

$$\Rightarrow \int_{\Omega} F = \int_{\Omega} x^2 + y^2 d\lambda_3(x, y, z) \stackrel{\text{válcové souřadnice}}{=} \int_1^2 \int_0^{2\pi} \int_0^z r^2 r dr d\varphi dz$$

$$= \int_1^2 2\pi \frac{z^4}{4} dz = \frac{\pi}{2} \left[\frac{z^5}{5} \right]_1^2 = \frac{\pi}{10} (32 - 1)$$

$$= \frac{31\pi}{10}.$$

integrirajeme ličnu funkcijo pri
(interval simetričnega okoli 0).

⑤ a)

$$a_2 = \frac{1}{\pi} \int_{-\pi}^{\pi} -t \cos(2t) dt \stackrel{\text{per-parite}}{=} 0 \quad \text{pro } 2 \in \mathbb{N}_0.$$

$$b_2 = \frac{1}{\pi} \int_{-\pi}^{\pi} -t \sin(2t) dt = \frac{2}{\pi} \int_0^{\pi} \underbrace{-t \sin(2t)}_{t \left(\frac{\cos(2t)}{2} \right)'} dt \stackrel{\text{per-parite}}{=}$$

$$= \frac{2}{\pi} \left\{ \left[t \frac{\cos(2t)}{2} \right]_0^{\pi} - \int_0^{\pi} \frac{\cos(2t)}{2} dt \right\}$$

$$= \frac{2(-1)^2}{2} - \frac{2}{\pi 2^2} [\sin(2t)]_0^{\pi} = \frac{2(-1)^2}{2} \quad \text{pro } 2 \in \mathbb{N}.$$

Fourierova vrsta funkcije f je

$$\sum_{k=1}^{\infty} \frac{2(-1)^k}{2} \sin(kt)$$

b) Poiščite $F_f(t)$ Fourierovo vrsto funkcije f na intervalu $[5\pi, 7\pi)$ je

$$F_f(t) = F_f(\underbrace{t-6\pi}_{\in [-\pi, \pi)}) = \begin{cases} -t + 6\pi, & t \in (5\pi, 7\pi), \\ 0, & t = 5\pi. \end{cases}$$